SQL Study Plan Day 5

Interview Questions

1. How would you design a stored procedure in MySQL to transfer money between two accounts while ensuring atomicity and consistency if one of the updates fails?

2. Write a stored procedure that accepts a department ID and returns the top three highest-paid employees, including ties. How would you handle invalid department IDs and departments without employees?

3. How would you implement error handling in a MySQL stored procedure using DECLARE HANDLER, SIGNAL, and RESIGNAL? Explain a scenario where each is required.

4. A stored procedure performs multiple INSERT, UPDATE, and DELETE operations across several tables. How would you ensure that a failure in any operation rolls back all changes made by the procedure?

5. How would you prevent SQL injection when building dynamic SQL inside a MySQL stored procedure? What limitations apply to parameter binding with PREPARE and EXECUTE?

6. A query filters a large Orders table using CustomerID, OrderDate, and Status. How would you determine the optimal composite index and column order? Explain how the leftmost-prefix rule affects your decision.

7. A table has separate indexes on DepartmentID and Salary. A query filters by both columns and sorts by Salary in descending order. How would you determine whether a composite index would outperform the existing indexes?

8. How would you investigate a query that performs a full table scan despite having an index on the filtered column? Which MySQL tools and execution-plan details would you inspect?

9. A frequently updated table contains several indexes, but write performance is degrading. How would you identify redundant or unused indexes and decide which ones to remove without harming critical queries?

10. How do covering indexes work in MySQL, and under what conditions can they eliminate additional table-row lookups? How would you evaluate their storage and write-performance trade-offs?

11. Two transactions update the same account rows in opposite orders and frequently encounter deadlocks. How would you diagnose the deadlocks and redesign the transactions to reduce their occurrence?

12. Explain the differences between READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, and SERIALIZABLE in MySQL InnoDB. How can these isolation levels affect non-repeatable reads, phantom reads, and locking behavior?

13. A transaction reads a product's available stock, checks whether sufficient stock exists, and then updates the quantity. How would you prevent concurrent transactions from selling more stock than is available?

14. What happens when a transaction performs DML operations, executes DDL statements, and then encounters an error? How do implicit commits and MySQL transaction boundaries affect rollback behavior?

15. How would you troubleshoot a transaction that remains open for a long time and causes lock waits, blocking queries, or growing undo history? Which MySQL monitoring commands or performance-schema tables would you inspect?

16. A database table stores customer details, multiple phone numbers, and multiple addresses in a single row using comma-separated values. How would you normalize the schema, define primary and foreign keys, and enforce referential integrity?

17. Given a table containing OrderID, CustomerID, CustomerName, ProductID, ProductName, Quantity, and UnitPrice, explain how you would decompose it into normalized tables up to Third Normal Form (3NF). Identify the functional dependencies and anomalies in the original design.

18. How would you determine whether a relational schema violates Second Normal Form (2NF) or Third Normal Form (3NF) when it uses a composite primary key? Provide an example involving partial and transitive dependencies.

19. A highly normalized database requires many joins and has become a performance bottleneck for reporting queries. How would you decide whether to denormalize the schema, and how would you maintain consistency when redundant data is introduced?

20. How would you design a database schema for an e-commerce system that must preserve historical product prices, customer addresses, and order details even when the original product or customer records change? Explain your normalization and referential-integrity decisions.

MCQ Question

1. In MySQL 8.0, which statement about transaction handling inside a stored procedure is correct?

A. Every stored procedure automatically starts a transaction and commits it when execution finishes.

B. A stored procedure can use START TRANSACTION, but transaction-control behavior depends on the surrounding transaction context and statements executed.

C. Every SQL exception automatically rolls back the entire transaction, regardless of any handler.

D. COMMIT inside a stored procedure affects only changes made by that procedure.

2. Which statement about MySQL composite indexes is correct?

A. An index on (CustomerID, OrderDate, Status) can efficiently support every query filtering only on Status.

B. An index on (CustomerID, OrderDate, Status) can generally support lookups using CustomerID alone because it is the leftmost column.

C. The order of columns in a composite index does not affect query performance.

D. MySQL always uses every column of a composite index when any one indexed column appears in a WHERE clause.

3. What is the primary purpose of EXPLAIN ANALYZE in MySQL 8.0?

A. To display only the optimizer's estimated execution plan without running the statement.

B. To execute the statement and report actual execution details alongside plan estimates.

C. To create indexes automatically based on query predicates.

D. To display only the indexes available on each table.

4. In InnoDB, what is a common cause of a deadlock?

A. Two transactions read the same row without acquiring conflicting locks.

B. Two transactions acquire locks in different orders and each waits for a lock held by the other.

C. A transaction executes a SELECT statement against a table with no indexes.

D. A transaction uses a consistent snapshot under REPEATABLE READ.

5. Under InnoDB's default REPEATABLE READ isolation level, what generally happens when a transaction performs multiple ordinary, non-locking SELECT statements within the same transaction?

A. Each SELECT necessarily sees the latest committed version of every row.

B. Each SELECT generally sees the same consistent snapshot, established by the first consistent read.

C. Every SELECT acquires exclusive locks on all rows it reads.

D. The isolation level prevents all deadlocks.

6. Which statement about MySQL error handlers in stored programs is correct?

A. A CONTINUE handler always terminates the stored procedure.

B. An EXIT handler terminates the entire MySQL server when an error occurs.

C. A CONTINUE handler resumes execution after the statement that raised the handled condition, while an EXIT handler exits the associated BEGIN...END block.

D. A handler can be declared only for duplicate-key errors.

7. Which of the following best describes a covering index?

A. An index that includes all columns required by a query, potentially allowing the query to avoid additional table-row lookups.

B. An index that automatically includes every column in its table.

C. An index that prevents duplicate values regardless of its definition.

D. An index that guarantees the query will never require sorting.

8. Which condition most directly indicates a violation of Third Normal Form (3NF)?

A. A table contains a composite primary key.

B. A non-key attribute depends on the whole candidate key.

C. A non-key attribute depends transitively on a candidate key through another non-key attribute.

D. A table contains a foreign key referencing another table.

9. Which statement about implicit commits in MySQL is correct?

A. Every SELECT statement causes an implicit commit.

B. Most DDL statements, such as CREATE TABLE and ALTER TABLE, cause implicit commits.

C. ROLLBACK always reverses DDL statements executed earlier in the transaction.

D. SAVEPOINT prevents implicit commits caused by DDL statements.

10. A transaction must decrement inventory only when sufficient stock is available, even under concurrent requests. Which approach is generally the safest?

A. Read the stock in an ordinary SELECT, check it in application code, and issue an unconditional UPDATE later.

B. Use a conditional atomic UPDATE that checks available stock in the WHERE clause, then verify the affected-row count.

C. Disable transaction isolation and allow concurrent updates.

D. Create a non-unique index on the stock quantity column and perform the check separately.

Puzzles

1. You have 8-, 5- and 3-litre jugs. Starting with the 8-litre jug full, how can you divide the water into two 4-litre portions?

2. Ten bags contain coins; genuine coins weigh 10 g and one bag has 9 g coins. Using one digital weighing, identify the faulty bag.

3. Cards show A, D, 4 and 7. Test the rule 'if a card has a vowel on one side, it has an even number on the other.' Which cards must be turned over?

4. Eight balls include one heavier ball. What is the minimum number of balance weighings needed to guarantee finding it? Explain a strategy.

5. One of 1,000 bottles is poisoned. With 10 test subjects and one testing round, how can the bottle be identified?

Aptitude Questions

1. The average of 15 numbers is 28. If 5 have average 24 and another 6 have average 30, what is the average of the remaining 4?

A) 28 B) 29 C) 30 D) 31

2. A batsman's average after 10 innings is 42. What must he score in the 11th innings to raise the average to 45?

A) 72 B) 75 C) 78 D) 80

3. The average of 8 consecutive even numbers is 31. What is the largest number?

A) 36 B) 37 C) 38 D) 39

4. A can do a job in 10 days and B in 15 days. They work together for 3 days. What fraction of the work is left?

A) 1/3 B) 1/2 C) 3/5 D) 2/5

5. A, B and C can complete a task in 24, 36 and 72 days respectively. If all three work together, how many days are needed?

A) 10 B) 12 C) 14 D) 16

6. A and B together complete a job in 12 days. A alone takes 18 days. B works alone for 6 days and then A joins. How many more days are needed after A joins?

A) 8 B) 9 C) 10 D) 12

7. A car travels 150 km. The first 60 km is at 30 km/h and the remaining 90 km at 45 km/h. What is the total travel time?

A) 4 h B) 4.5 h C) 5 h D) 5.5 h

8. A train 200 m long crosses a pole in 12 seconds. How long will it take to cross a 300 m platform at the same speed?

A) 24 s B) 30 s C) 36 s D) 40 s

9. A boat covers 24 km downstream in 2 hours and the same distance upstream in 3 hours. What is the speed of the stream?

A) 1 km/h B) 2 km/h C) 3 km/h D) 4 km/h

10. How many ways can the letters of APPLE be arranged?

A) 60 B) 90 C) 120 D) 180